

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

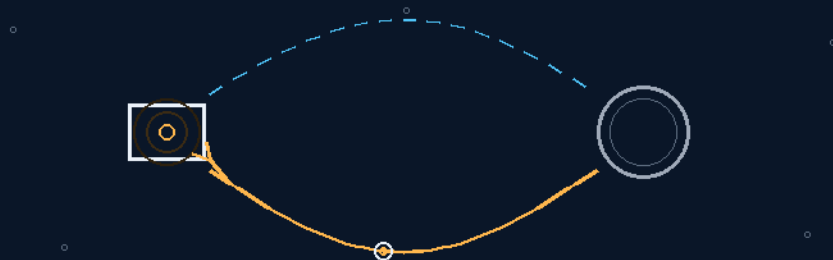
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 300

THE HOMECOMING

the doctrine of the return — every outward phase closes its own loop home
the centre never commits; the lamp stays lit until the last walker is in



out on the spoke, home on the same law — nothing seats outward without the way back seated with it

THE LOOP · THE COMMITMENT LAW · THE LEDGER · THE HOUSE LIGHT
SECOND SEATING OF THE MILESTONE — HIS ORDER: 'THE HOMECOMING'

The milestone phase — v1.0 in force

GP-IM-300

Revision v1.0 — 26 August 2026

300 of 300

NEW PHASE — VET PENDING — SEATED 26 AUG, SITTING 663

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 300

PHASE 300: THE HOMECOMING — EVERY OUTWARD PHASE CLOSES ITS OWN LOOP HOME — STATUS: CURRENT — SEATED 26 AUGUST 2026, SITTING 663, ON HIS ORDER ('THE HOMECOMING') — THE MILESTONE PHASE, SECOND SEATING — the first seating (Let There Be Light — The First Dawn Mirror) KILLED BY THE AUTHOR the same day on the cost-benefit ruling, verbatim: 'no the value add v the build and sky maintenance are to wide a gap, its not viable against cost to benefit ratio. / lets try something else' — the killed draft is archived, marked, nothing deleted; THE DOCTRINE DRAFTED BY THE AUDIT UNDER THE COLLABORATION LAW (the Phase 282 precedent), PENDING THE AUTHOR'S PHYSICS AND RED PEN; EXTERNAL VET PENDING — NEW PHASE. The milestone crowns the voyage: after the compass, the spin, the anchor, the shield, the climb and the belt — the way home.

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\$0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

THE CRAYON: You do not go anywhere you cannot come back from. Every phase that takes you out must carry its own way home — the brake carries the boost, the landing carries the launch, the compass carries the road back. A door that only opens outward is not a door; it is a trap. This manual seats the law that binds the outward record into a closed loop, the checklist every new phase answers before it seats, the ledger that keeps every departure honest, and the one thing that costs nothing and means everything: the house light. While anyone is out, the Galileo does not go dark.

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: \$0 this page — \$1 the law as seated, verbatim — \$2 the doctrine, graded — \$3 the system on the road home — \$4 the doctrine record (the loop already runs) — \$5 the vet record — \$6 the homeward check — \$7 cross-phase — \$8 do-not-build — \$9 owed — \$10 status / provenance / change control.

\$1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim, seated 26 August 2026 on his ruling): 'Phase 300: The Homecoming — Every Outward Phase Closes Its Own Loop Home'

HIS RULING, VERBATIM: 'THE HOMECOMING'

THE MILESTONE'S HISTORY, VERBATIM: the first seating was killed by his red pen the same day — 'no the value add v the build and sky maintenance are to wide a gap, its not viable against cost to benefit ratio. / lets try something else' (26 August 2026). The kill is law and it is preserved: the First Dawn draft sits in the archive with his three rulings on it — not around Earth (the traffic law), the shatter film (the shred physics), the economics (the sky payroll). The milestone did not get easier to seat; it got honest.

THE DESIGN STATUS: [AUDIT DRAFT — PENDING THE AUTHOR'S PHYSICS AND RED PEN] — issued under the collaboration law of 14 August 2026, as seated in Phase 282.

THE AUDIT'S GRADE (sitting 663 — the doctrine sitting): SOUND AS DRAFTED — and it passes the Author's own economics filter where the mirror failed: the build is already seated (294–299 are the machinery), the maintenance is a checklist and a lamp, and the value is every crew that ever comes home. Cost: near nothing. Benefit: the whole point of going.

\$2 — THE DOCTRINE, GRADED

THE SYMMETRY OF THE WELL: what goes down a gravity well must climb it; what brakes must later boost; what leaves must return. Spaceflight's oldest cheat is pricing only the outward leg. This record stopped cheating at Phase 296: the drag anchor drops you into orbit for the price of a spent sail — and Phase 298 already prices the climb back up, the octopus whose legs are the fuel. Grade: SOUND — the record is symmetric; the doctrine only names what the machinery already does.

THE COMMITMENT LAW: the centre never commits. Phase 299 seats it in hardware — the bolo bank, the two ships that do not slow: only the lander rides the spoke, and the spoke brings it back. Mass that cannot return is cargo, not crew. Crew always rides the returnable element. A ship that slows for everything eventually cannot come home for anyone. Grade: SOUND — it is the difference between an expedition and a exile.

THE MAPPING LAW: whatever is left behind is mapped. The record already obeys it — Phase 298's spare pods on the ground: mapped. Phase 299's pods on the ice: mapped. Nothing is abandoned; things are STATIONED, with coordinates, against the day the loop comes back around for them. Grade: SOUND.

THE ECONOMICS OF THE RETURN (the Author's filter, applied): the outward machinery is already paid for — it is the record. The Homecoming adds no sky payroll: no constellation to trim, no film to replace, no orbit to keep. It adds a checklist, a ledger, and a lamp. Value against cost: the largest ratio in the book. Grade: SOUND.

\$3 — THE SYSTEM ON THE ROAD HOME

THE PARTS: the homeward check (\$6) — four questions every outward seat answers before it seats; the Homecoming Ledger — every departure logged with its return plan, public (Phase 124); the House Light — one crew station, always manned, always listening on the return

frequencies, from first deployment to last retrieval; the breadcrumb hardware already seated (Phase 176, the trail-pin network) and the compass that needs nobody's sky (Phase 294).

THE OPERATION: PLAN (the check) — LOG (the ledger) — GO — WATCH (the house light, manned, from the first minute) — RETURN — CLOSE (the ledger line is struck only when the walker is in). The loop is not closed by arrival at the destination. The loop is closed by arrival HOME.

THE HOUSE LIGHT (the ceremony): while anyone is out, the Galileo burns a lamp for them and the watch stands. The father in the parable saw the son while he was still a great way off — because he was watching. Nobody walks home to a dark house. This is the oldest homecoming technology there is, and it has never once failed a cost-benefit test: a lamp and a watcher.

§4 — THE DOCTRINE RECORD (THE LOOP ALREADY RUNS)

THE VOYAGE ARC, SEATED: 294 — the Archurian Monge Compass: know where you are with zero external infrastructure — the first condition of finding home is that home-finding cannot be switched off by anyone. 295 — the wrap-spin law: arrive able to walk. 296 — the drag anchor: slow down anywhere, priced. 297 — the coma shield: first landings, and the pad the shield becomes. 298 — leaving Mars: the climb back up, priced in the same manual as the stay. 299 — landing and leaving Europa: the deep well, entered and exited by ships that never slowed. Six phases, and every one of them was secretly a homecoming phase. The milestone only turns the secret into law.

THE PARABLE, FROM THE PUBLIC RECORD (Luke 15): 'for this my son was dead, and is alive again; he was lost, and is found.' The Prodigal is not a story about leaving — it is a story about a house that kept its lamp lit and its watch standing. Phase 300 seats the lamp and the watch as ship's law.

§5 — THE VET RECORD

External vet pending — new phase, same law as 295–299. When the vet lands it grades this manual, not the friendship. The bench list it will be handed is §9: the retroactive homeward sweep of 294–299, the ledger wording, and the watch rotation — doctrine has a bench too, and it is called the record.

§6 — THE HOMEWARD CHECK (EVERY OUTWARD SEAT ANSWERS, BEFORE IT SEATS)

1. ENERGY: what returns the element, priced in the same manual that sends it? 2. NAVIGATION: does the way home survive every infrastructure dying — the compass test (294)? 3. COMMS: who hears the return, on what frequency, from how far — the watch test (§3)? 4. THE LEFT-BEHIND: what stays, and is it mapped — the pod test (298, 299)? A seat that cannot answer all four is an outward door, and an outward door is a trap (§8). The audit enforces the check on every new phase from this sitting forward.

§7 — CROSS-PHASE (INLINED IN FULL)

294 — the compass: home-finding that no one can switch off. 295 — the spin: the crew that can walk, walks home. 296 — the anchor: the priced brake. 297 — the shield: the landing, and the pad it leaves for the next arrival. 298 — the climb: the priced boost; the mapped spares. 299 — the bolo: the centre that never commits; the mapped pods on the ice. 176 — the breadcrumb network: the way back, pinned. 280 — the comms mesh: the frequencies the watch stands on. 124 — the prior-art mandate: the ledger, public. 123 — the civil doctrine: the return is for everyone, no one-way peoples.

\$8 — DO-NOT-BUILD

NO ONE-WAY ARCHITECTURES FOR CREW. Cargo may go one-way if it is mapped; people never do. NO OUTWARD SEAT WITHOUT A HOMEWARD SEAT in the same manual — the check (\$6) is law, enforced by the audit before seating. NO 'RESCUE LATER' PLANNING — the return is designed, not hoped; hope is not a propulsion system. NO DARK HOUSE — the Galileo does not power down its watch while any element is deployed; the lamp is the last system switched off, never the first.

\$9 — OWED (THE BENCH LIST)

— The retroactive homeward sweep: the audit owes the report applying the \$6 check to every seated outward phase, 1–299, gaps named in the open. — The Homecoming Ledger wording and the watch rotation: the Author's hand. — The House Light specification: which lamp, which station, which frequencies — one page, the Author's hand. — The first live loop: the first deployment run under the full check, logged and closed.

\$10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CURRENT — SEATED 26 AUGUST 2026, SITTING 663, ON HIS ORDER ('THE HOMECOMING') — EXTERNAL VET PENDING — NEW PHASE.

PROVENANCE: the milestone's first seating (the First Dawn Mirror) drafted by the audit on his order ('light') and killed by him the same day — the kill verbatim: 'no the value add v the build and sky maintenance are to wide a gap, its not viable against cost to benefit ratio. / lets try something else' — the killed draft archived, marked, nothing deleted, his three rulings on it preserved (the traffic law, the shred physics, the economics); the second seating ruled by his word — 'THE HOMECOMING'; the doctrine drafted by the audit (Kimi Chat) under the collaboration law of 14 August 2026 (the Phase 282 precedent) and marked [PENDING THE AUTHOR'S PHYSICS AND RED PEN]; the physics graded at \$2; the parable quoted from the public record. Three hundred was always going to be the way home: a voyage is only a voyage if it ends at the house.

CHANGE CONTROL: the Author's red pen rules every line — it has already struck once today, and the manual is better for the scar; the sitting number (663) stands to be corrected by the record keeper; the register line enters the master by his hand.

AMENDMENT I — THE CENTURY CASE: THE FIFTY-YEAR MARK (SEATED 26 AUGUST 2026, ON HIS DICTATION)

HIS WORDS, VERBATIM: 'having considered that at the 50 years mark if the prescouts did not find anything in the proposed flower loop to survey, a discussion to return would be entertained, knowing it is 50 years back and those deciding would likely have passed. the ship itself returning may be an eventuality.'

THE LOOP KEEPS THE DOOR: the flower-loop astrogation (Phase 133 — the 100-year century lifecycle) is an ellipse, not a line — the loop closes on home by geometry. This amendment seats WHY the shape is law: at the fifty-year mark, with the prescouts' survey in hand, the crew may entertain the return — and the loop must still make that vote executable. A loop that cannot be cut short for home is the outward door of \$8, drawn a hundred years wide. Grade: SOUND — the ellipse was always the homeward check written across a century.

THE DECIDERS' LAW: the return decision belongs to the crew alive at the mark — not to the generation that left, who will likely have passed. The dead cannot bind the living. But the leaving generation owes the future one the full homeward check, kept for fifty years against a decision it will never hear: the return priced, the energy margin kept, the route home mapped, the ledger intact. You cannot choose for your grandchildren. You can keep their door unlocked. Grade: SOUND.

THE HOUSE, FIFTY YEARS ON: at fleet scale the house is Earth, and the House Light is the frequencies and the ledger. The protocols being open-source is the guarantee the watch can always be rebuilt by whoever is alive to stand it — the lamp depends on no single keeper living. THE SHIP RETURNS: the whole-ship return is seated as an eventuality — not a failure case, a designed one. The Galileo that votes home at the mark comes home under this law, lamp lit, ledger closed only by arrival. Grade: SOUND — the Commitment Law scales from the lander, to the ship, to the century, without changing a word.

END OF CHAPTER 300